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Boolean Universality and the Null Cone of the Tetrad Frame

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Abstract
Four null directions of a Lorentzian frame that single out no spatial direction form a regular tetrahedron; with their antipodes they are eight null directions, the corners of a cube, labelled canonically (up to the cube's 48 symmetries) by the eight three-bit words. It has been claimed that this sampled null cone is a universal Boolean operator. We separate what is generic from what depends on the null cone. Generically, for any Boolean cube: the Walsh–Hadamard matrix represents every function, but parity gates, including the split into the two tetrahedra, generate only the 16 affine functions and are not functionally complete; and $H_8 = S\,D\,S^T$, where S is the table of AND monomials and D carries a phase $e^{i\pi}$ per set bit. Over GF(2), H_8 is the all-ones matrix while S is invertible, so the algebraic normal form, and with it functional completeness, lives in S. The null cone enters through the two-bit block of the frame: det is the unique quadratic form invariant under independent changes of basis of the two bits, so the null cone there is canonical, and a two-input gate is null exactly when its truth table is separable. Ten gates are null; they include AND and NOT, and NOR, a real light ray, is universal on its own, while parity is off the cone. Read mod 2, the null rotation of the frame sends each gate to its algebraic normal form.

Keywords: null cone, tetrad, stellated octahedron, Boolean functions, functional completeness, Walsh–Hadamard transform, algebraic normal form, null rotation, Segre variety

1 Introduction

Letter 1 of the tetrad series derives the stellated octahedron from the null cone [1]: four null directions that introduce no preferred spatial direction form a regular tetrahedron on the celestial sphere, and with their antipodes they are the eight points of the star. The research archive then states, as a claim, that this sampled null cone is a universal Boolean operator [2]. The claim rests on two facts: the cube's eight corners carry every three-input Boolean function, and a coupled oscillator laid out on them, the octahedral resonance node, computes with the Walsh spectrum.

"Universal" has two meanings, and they come apart here. A set of points is universal in the weak sense if every Boolean function on it can be represented; any eight labelled points are. A set of gates is functionally complete if composing it with the projections produces every Boolean function; constants count only if they belong to the set. Post's theorem [5] says a set is complete exactly when it lies in none of five classes: the functions preserving 0, those preserving 1, the monotone, the self-dual and the affine.

Contributions and prior work. Theorems 1 and 2 are facts about any labelled Boolean cube. Post's theorem, the algebraic normal form [6, 7, 8] and the completeness of NOR [9] are classical, and the identity $H_8 = S\,D\,S^T$ is elementary; we claim no novelty for any of them. What this paper adds is their placement on the frame, and two points where the null cone itself is used. First, the three-bit labelling of the null directions is canonical (Proposition 1). Second, on the two-bit block the Minkowski form is forced (Lemma 1), and its null vectors are exactly the separable gates (Theorem 3), which are universal while parity is not. The result is a precise version of the archive's claim, with the generic and the geometric parts kept apart. Figure 1 summarises it.

2 The Eight Null Directions

Proposition 1. *In a Lorentzian frame, let $u_1, \dots, u_4$ be unit spatial vectors with zero centroid and isotropic second moment, so that each $k_I = (1, u_I)$ is null. Then $u_I \cdot u_J = -\frac{1}{3}$, and up to rotation the u_I are $v/\sqrt{3}$ for the four corners $v \in \{\pm 1\}^3$ with $v_1 v_2 v_3 = +1$ (TetA). With their antipodes (TetB) they are the cube's eight corners. Writing $b_i = (1 - v_i)/2$, the eight null directions are the eight words $b \in \{0,1\}^3$, and TetA and TetB are the words of even and odd weight.*

Proof. Squaring the centroid condition gives $4 + 12\,u_I \cdot u_J = 0$, and four unit vectors with equal pairwise products $-\frac{1}{3}$ are a regular tetrahedron, unique up to $O(3)$ [1]. Under $v_i = (-1)^{b_i}$, $v_1 v_2 v_3 = (-1)^{|b|}$. □

Remark (the labelling). The rotation in Proposition 1 is fixed up to the cube's symmetry group, of order 48, which acts on the words by

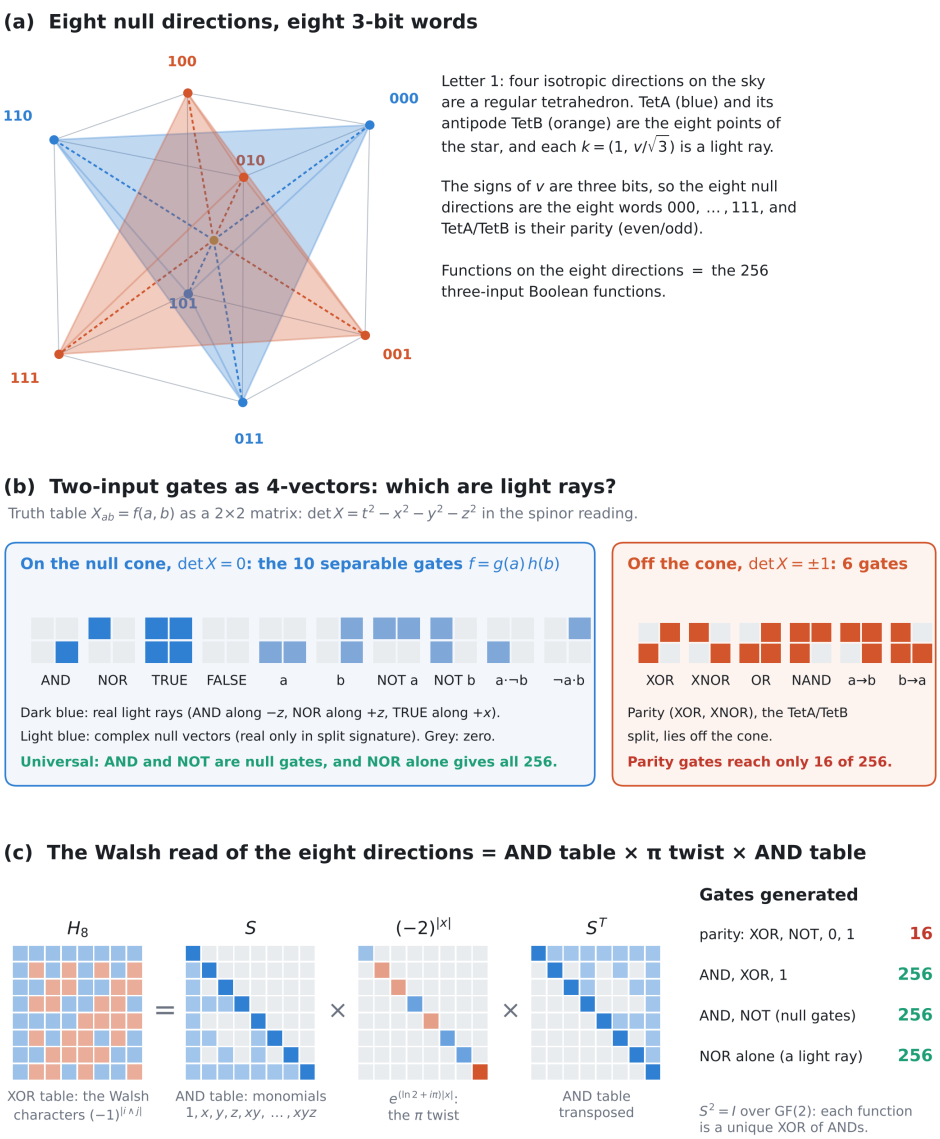


Figure 1. (a) The eight null directions of the tetrad frame (Proposition 1), labelled by the signs of their spatial parts: the eight three-bit words, with the two tetrahedra the even and odd words. (b) The 16 two-input gates on the two-bit block, with truth tables read as 2×2 matrices X, for which $\det X$ is the Minkowski norm (Lemma 1). Ten gates are null: the separable gates $g(a)h(b)$. AND, NOR and TRUE are real light rays; the other six, including parity, are off the cone (Theorem 3). (c) The Walsh read of the eight directions factors as the AND table, a phase $e^{i\pi}$ per set bit, and the transposed AND table (Theorem 2). Right: the number of three-input functions each gate set generates.

permuting and negating bits. Every property used below is invariant under these 48 relabellings: the affine class, the gate closures, and separability. An odd number of negations exchanges TetA and TetB. Letter 1 writes the signature as $(-,+,+,+)$; below we use $(+,-,-,-)$, so that the norm is $\det X$. The light cone is the same.

So a function on the eight null directions is a three-input Boolean function, and there are 256 of them.

3 Representation Versus Gates

Let H_8 have entries $(-1)^{|i\wedge j|}$, the Walsh characters of the cube [10, 11, 12], where $i \wedge j$ is the bitwise AND of the words and $|\cdot|$ counts ones.

Proposition 2 (representation). $H_8^2 = 8I$, *so every function on the eight null directions is a unique real combination of the eight Walsh characters.*

This is the sense of the archive's claim. It is generic: the same holds for any eight labelled points, and it says nothing about which gates the structure supplies. In 0/1 form the Walsh characters are the parities $x_J = \bigoplus_{i\in J} b_i$.

Theorem 1 (parity is not universal). *Every Walsh character, and the TetA/TetB*

indicator, is affine over GF(2): its algebraic normal form has degree at most one. Closing the three projections under XOR, NOT and the constants gives exactly the 16 *affine functions of three bits, not all* 256*. So no set of parity gates is functionally complete.*

Proof. A parity is a sum of variables over GF(2), and a composition of affine functions is affine. The count of 16 is checked by exhaustive closure. By Post's theorem, a gate set inside the affine class is not complete. □

As the archive describes the octahedral node [2], its XOR is a grade-two Walsh coefficient and its parity a grade-three one; both are affine. Its NAND and full adder are read as a dot product with the gate's Walsh coefficients, that is, through Proposition 2. We make no independent claim about the node.

4 The Walsh Read Factors Into AND and XOR

Let $S_{ij} = 1$ when the bits of j are a subset of those of i, and 0 otherwise. Equivalently $S = A^{\otimes 3}$ with $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, the Sierpiński triangle.

Theorem 2. *(i)* S_{ij} *is the AND of the bits of* j *at the corner* i*:* S *is the evaluation table of the eight AND monomials* $1, x, y, z, xy, xz, yz, xyz$*. (ii)* $S^2 = I$ *over GF(2), so every Boolean function is a unique XOR of AND monomials, its algebraic normal form, with coefficients* $Sf \bmod 2$*. (iii) With* $|x|$ *the number of set bits of the index* x*,*

$$H_8 = S\,D\,S^T, \qquad D = \operatorname{diag}\big((-2)^{|x|}\big) = \operatorname{diag}\big(2^{|x|} e^{i\pi|x|}\big). \tag{1}$$

(iv) AND, XOR and the constant 1 *generate all* 256 *functions. (v) Over GF(2),* $H_8 \equiv J$*, the all-ones matrix, of rank* 1*, while* S *is invertible.*

Proof. (i) The monomial $\prod_{b\in J} x_b$ is 1 at a corner exactly when every bit of J is set there. (ii) $A^2 \equiv I \pmod 2$, and the same holds for tensor powers; this is the Zhegalkin or Reed–Muller expansion. (iii) Entry by entry, with $n = |i \wedge j|$, $\sum_{K\subseteq i\wedge j} (-2)^{|K|} = (1-2)^n = (-1)^n$. (iv) follows from (ii), and is also checked by exhaustive closure. (v) Every entry of H_8 is ± 1. □

Part (v) is the precise sense in which universality is located in the Sierpiński factor. Mod 2 the Hadamard sign pattern carries no information, while S carries the whole algebraic normal form. The factor $e^{i\pi|x|}$ is a phase of -1 per set bit. Its reading as a rotation of space exists only on the two-bit block, where $A \otimes A$ is a null rotation and $H_2 \otimes H_2$ a rotation by π [4].

5 The Null Cone of the Two-Bit Block

This section concerns a different object from Section 2. There the null directions were the inputs, labelled by three-bit words. Here the object is the two-bit block $\mathbb{C}^2 \otimes \mathbb{C}^2$ of the frame, the four-dimensional factor in $H_8 = H_2 \otimes H_4$ [3]. Its vectors are functions of two bits, so a two-input gate f is the vector $X_{ab} = f(a, b)$, a 2×2 matrix.

Lemma 1 (the form is forced). *Up to scale, the only symmetric bilinear form on* $\mathbb{C}^2 \otimes \mathbb{C}^2$ *invariant under independent* $SL(2)$ *changes of basis of the two bits is the polarisation of* $\det X$*. Writing* $X = t + x\sigma_1 + y\sigma_2 + z\sigma_3$*,* $\det X = t^2 - x^2 - y^2 - z^2$*.*

Proof. The symmetric square of $\mathbb{C}^2 \otimes \mathbb{C}^2$ is $(\mathrm{Sym}^2 \otimes \mathrm{Sym}^2) \oplus (\Lambda^2 \otimes \Lambda^2)$, and only the last summand, spanned by $\varepsilon \otimes \varepsilon$, is invariant [13]. Solving the invariance equations directly gives a one-dimensional solution space. □

So the null cone of the two-bit block is not a choice. Its null vectors are the rank-one matrices, the Segre cone of product vectors, and this is invariant under every change of basis $X \mapsto BXC^T$. What does depend on a choice is which null vectors are real light rays: that needs a Hermitian structure. We use the standard one, in which the frame's legs form a Newman–Penrose null tetrad [4].

Theorem 3 (the null gates). *A gate is null,* $\det X = 0$*, exactly when* $f(a, b) = g(a)\,h(b)$*. Exactly* 10 *of the* 16 *gates are null: FALSE, TRUE, AND, NOR, the literals* a*,* b*,* $\neg a$*,* $\neg b$*, and* $a \wedge \neg b$*,* $\neg a \wedge b$*. The other six, XOR, XNOR, OR, NAND and the two implications, have* $\det X = \pm 1$*. This classification does not depend on the reading. In the standard Hermitian structure, AND, NOR and TRUE are real light rays along* $-z$*,* $+z$ *and* $+x$*; FALSE is the zero vector; and the other six null gates are complex null vectors. Re-reading one bit, for example by a swap, can make AND complex.*

Proof. A 2×2 matrix is singular exactly when its rank is at most one, and a 0/1 matrix of rank one is an outer product $g\,h^T$ of 0/1 vectors. Rank is unchanged by $X \mapsto BXC^T$. The components follow from $x^\mu = \frac{1}{2}\mathrm{Tr}(\sigma_\mu X)$; a non-symmetric X has an imaginary y component. □

Corollary. *The null gates are functionally complete, since they contain AND and NOT. NOR, a real light ray, is complete on its own. Parity, XOR and XNOR, lies off the cone.*

Proof. {AND, NOT} and {NOR} are classical complete sets; exhaustive closure confirms that each generates all 256 three-input functions. □

Proposition 3 (the null rotation mod 2). *The Sierpiński null rotation $X \mapsto AXA^T$ fixes the light ray $|11\rangle$, which is AND. Over the reals its only fixed gates are FALSE and AND. Mod 2 it sends each gate's truth table to the coefficients of its algebraic normal form, and the gates it fixes mod 2, those equal to their own normal form, are FALSE, AND, XOR and OR.*

Proof. A fixes $|1\rangle$, and $\mathrm{vec}(AXA^T) = (A \otimes A)\,\mathrm{vec}(X) = S_2\,\mathrm{vec}(X)$, which mod 2 is Theorem 2(ii) on two bits. The fixed gates are found by checking all 16. □

Rank-one truth tables are the combinatorial rectangles of communication complexity [14]. The observation here is that, under the forced metric of Lemma 1, they are the light rays of the frame's two-bit block.

6 Discussion and Scope

The archive's claim is true in two senses, and they should be kept apart. In the weak sense, representation, it holds for the eight null directions as it does for any eight labelled points (Proposition 2). In the strong sense, functional completeness, it holds in a specific way. Parity alone, the structure that separates TetA from TetB, is not complete (Theorem 1). The Walsh read factors through the AND table, which carries the whole algebraic normal form (Theorem 2). On the two-bit block the forced metric makes the light rays exactly the separable gates, which are complete, while parity is off the cone (Lemma 1, Theorem 3).

The paper does not show that physical light computes, or that a lattice of photons is a computing medium; those remain the archive's interpretation. The distinction between real and complex light rays depends on the Hermitian structure, though the null set does not. The ten null and six non-null gates should not be confused with the archive's split of 16 into ten symmetric and six antisymmetric components: the null gates do not form a linear subspace, and no relation between the two counts is claimed.

Every statement is checked by the accompanying script (26 checks, exact arithmetic, with closures exhaustive over all 256 functions).

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